Test 11 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

#	Benchmark	Answer	Explanation
1	MA.6.AR.1.1 moderate	the quotient of a number x and 6 → x/6; a number x decreased by 6 → x - 6; the sum of a number x and 6, divided by 2 → (x + 6)/2	Translate each phrase one operation at a time.
2	MA.6.DP.1.6 easy	30	The new value becomes the greatest value, so the range grows.
3	MA.6.NSO.3.5 easy	9/20	A percent means hundredths, so write it over 100 and simplify.
4	MA.6.AR.1.3 moderate	6x + 3 → 21; 6x^2 → 54; 3 - x → 0	Substitute the value of x, then follow the order of operations.
5	MA.6.NSO.2.2 moderate	3/4 × 1/2, 1/2 × 3/4, 3/8	Multiplication of fractions can be done in either order, and the product of the numerators over the product of the denominators is the same value.
6	MA.6.NSO.2.3 moderate	7.12 × 8, 8 × 7.12, 56.96 ÷ 1	The total is the price times the number of tickets; order does not matter.
7	MA.6.AR.3.4 easy	40	Divide the part by the percent written as a decimal to find the whole.
8	MA.6.NSO.1.4 moderate	-10, 9	A reading sets off the alarm when its absolute value is greater than 8.
9	MA.6.AR.2.2 moderate	15, 33	Undo addition with subtraction, and undo subtraction with addition.
10	MA.6.NSO.3.1 moderate	2, 3, 4, 6	A common factor divides both numbers with no remainder.
11	MA.6.AR.3.5 easy	0.5	Divide the price by the number of bottles to compare unit prices.
12	MA.6.DP.1.1 moderate	How many minutes did each sixth grader spend on the bus this morning?, How many books did each student in the class read last month?	Look for questions that many people would answer with different numbers.
13	MA.6.AR.2.4 moderate	2.82, 5.36	Undo the operation shown in each equation.

#	Benchmark	Answer	Explanation
14	MA.6.NSO.1.2 moderate	There is no money left on the card and no amount is owed.	In a balance context, 0 is the point between having credit and owing money.
15	MA.6.NSO.4.2 easy	-9	Divide the absolute values, then decide the sign from the two signs.
16	MA.6.AR.2.3 moderate	11, 55	Divide to undo multiplication; multiply to undo division.
17	MA.6.DP.1.2 moderate	Mean → 14; Median → 14 1/2; Range → 16	Find the mean by dividing the total, the median as the middle value, and the range as the greatest minus the least.
18	MA.6.DP.1.4 moderate	10	Count every dot on the plot, including repeats.
19	MA.6.AR.3.2 moderate	16 miles in 4 hours → 4; 16 dollars for 2 pounds → 8; 72 pages in 6 minutes → 12	Divide the first quantity by the second in each rate.
20	MA.6.GR.2.2 moderate	108, 12	Decompose the figure: whole rectangle minus the missing corner.
21	MA.6.AR.3.1 moderate	3, 2	Divide both quantities by their greatest common factor.
22	MA.6.AR.1.2 moderate	60.0	The boundary value is 60, the number the inequality compares x to.
23	MA.6.AR.2.1 moderate	8 → False; 10 → True; 11 → True; 19 → True	Values of 10 or greater make the inequality true.
24	MA.6.GR.1.1 moderate	A (3, 4) → Quadrant I; D (5, -2) → Quadrant IV; C (-4, -3) → Quadrant III	The signs of the coordinates tell which quadrant a point is in.
25	MA.6.GR.2.1 moderate	base 5 feet, height 8 feet → 20; base 9 feet, height 6 feet → 27; base 7 feet, height 4 feet → 14	Use one half times base times height for each triangle.
26	MA.6.NSO.3.4 easy	45	Multiply all of the prime factors together.
27	MA.6.DP.1.3 moderate	Minimum → 2; Median → 9; Upper quartile → 14	Read the five-number summary from left to right on the box plot.
28	MA.6.NSO.2.1 moderate	0.4 × 6, 24 ÷ 10, 1.2 × 2	Evaluate each expression and compare it to the target value.
29	MA.6.NSO.1.3 moderate	-6 1/2, 7	Compare how far each number is from 0, ignoring its sign.

#	Benchmark	Answer	Explanation
30	MA.6.NSO.3.2 moderate	12(11 + 7), 132 + 84, 12 × 18	Use the distributive property: the factored form must multiply back to the original sum.
31	MA.6.NSO.3.3 moderate	6^2, 6 × 6, 36 × 1	A power repeats the base as a factor; it is not the base times the exponent.
32	MA.6.NSO.3.4 moderate	2^2 × 3 × 11, 2 × 2 × 3 × 11, 132 × 1	Each correct expression multiplies back to the original number.
33	MA.6.AR.1.4 moderate	7(x + 3) → 7x + 21; 7x + 6x → 13x; 7x + 3 - 3 → 7x	Use the distributive property and combine like terms.
34	MA.6.NSO.4.1 easy	17	Subtracting a number is the same as adding its opposite.
35	MA.6.NSO.1.1 moderate	-2 1/2, -1 3/4	A number is less than -1 1/2 when it lies to the left of -1 1/2 on the number line.
36	MA.6.GR.2.4 moderate	40, 48	Each pair of opposite faces has the same area, so double the area of one.
37	MA.6.GR.1.3 moderate	28	Add the four side lengths, or use 2 times the sum of the length and width.
38	MA.6.GR.2.3 moderate	2 by 5 by 6 inches → 60; 2 by 2 by 9 inches → 36; 4 by 4 by 3 inches → 48	Multiply the three dimensions of each prism.